Hands-on 4 : Autoencoders and latent space representation

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DLMI2026

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Fonds de recherche Nature et technologies Québec

cnrs

itech santé

CIMS Centre d'imagerie moléculaire de Sherbrooke

CHUS Centre hospitalier universitaire de Sherbrooke

LABEX PRIMES UNIVERSITÉ DE LYON

MICCAI

ÉTS Le génie pour l'industrie

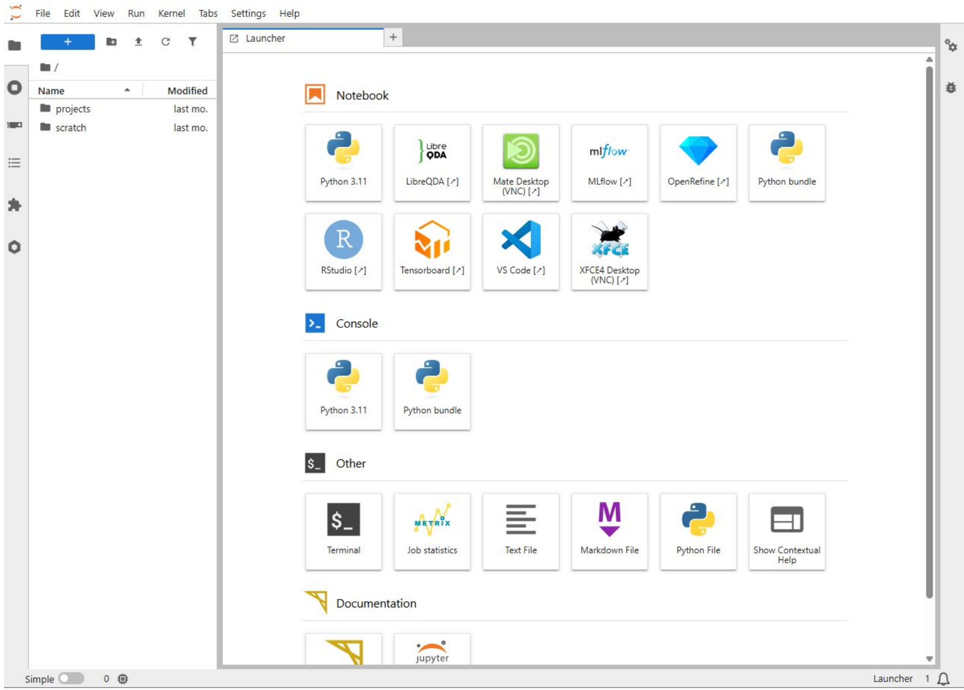
CREATIS

UNIVERSITÉ DE SHERBROOKE

Start a session

As for the previous hands on session, long onto the CaulculQuebec environment and start a compute node

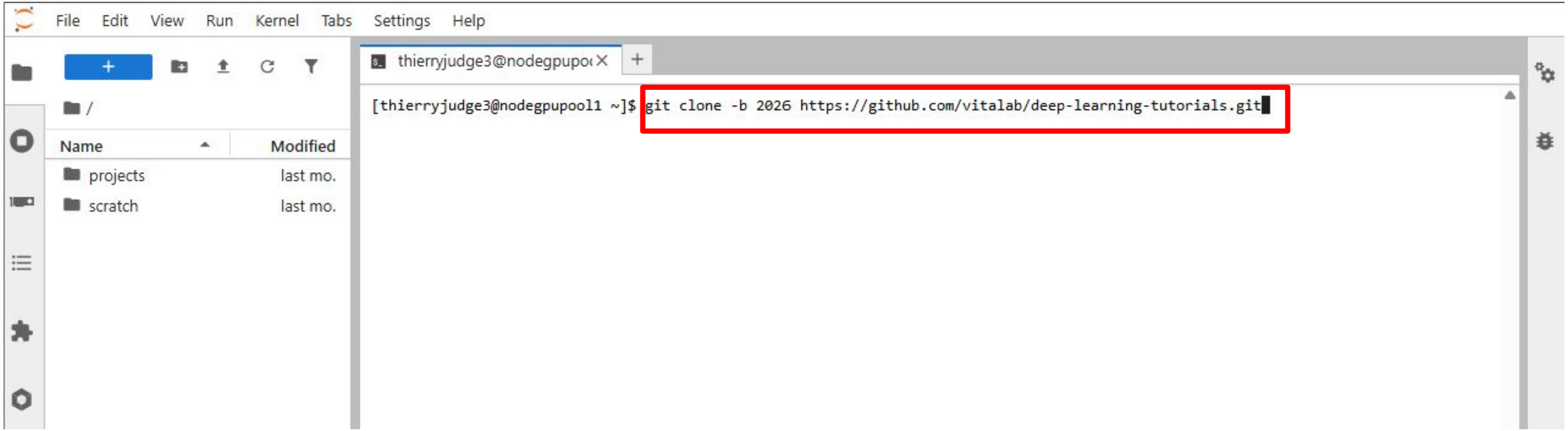
https://jupyter.dmli.calculquebec.cloud/



Clone the repository

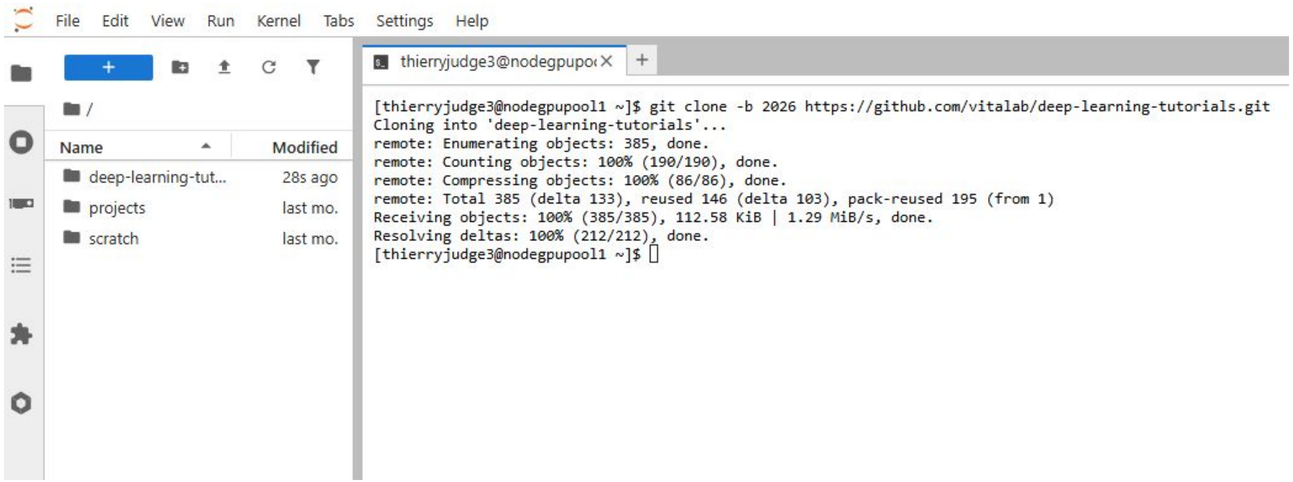
In the terminal, copy the following command and enter

git clone -b 2026 https://github.com/vitalab/deep-learning-tutorials.git



Clone the repository

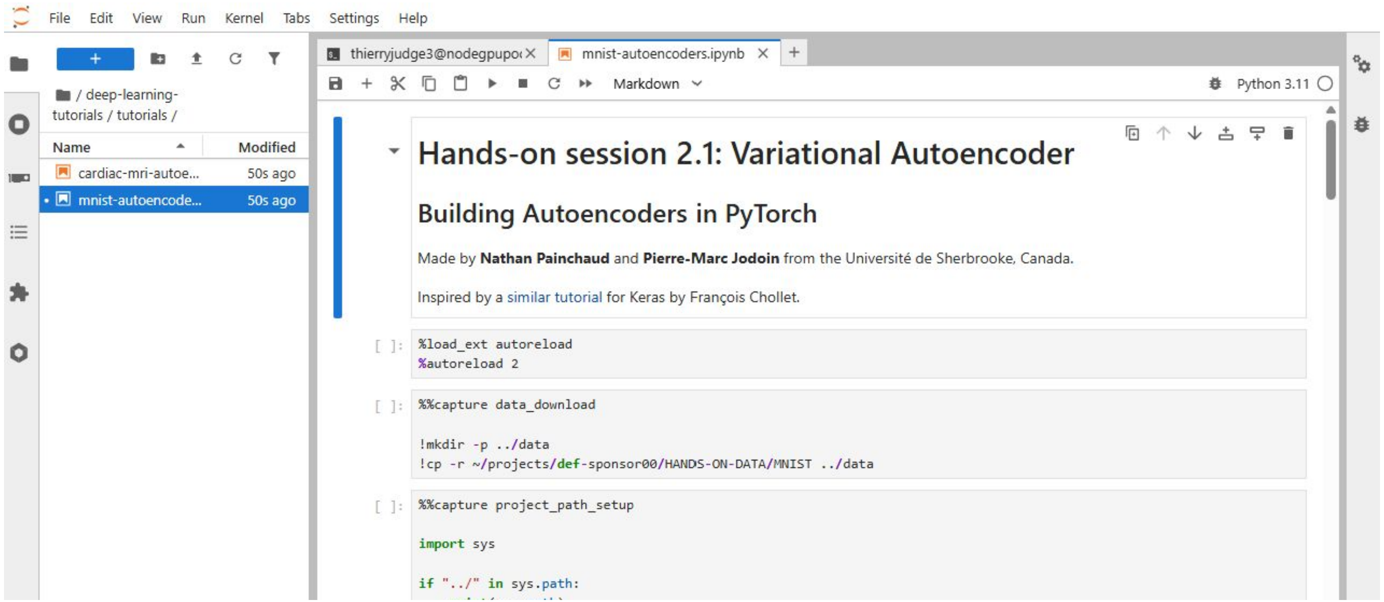
You should see the following output in the terminal and a new folder should appear on the left. If you do not see the folder click on refresh



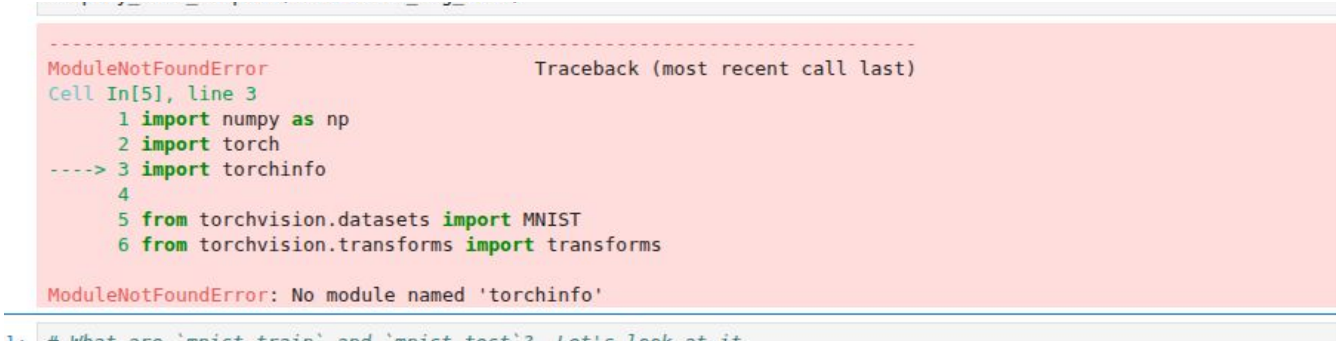
One the first notebook

Using the menu on the left, navigate to **deep-learning-tutorials/tutorials**.

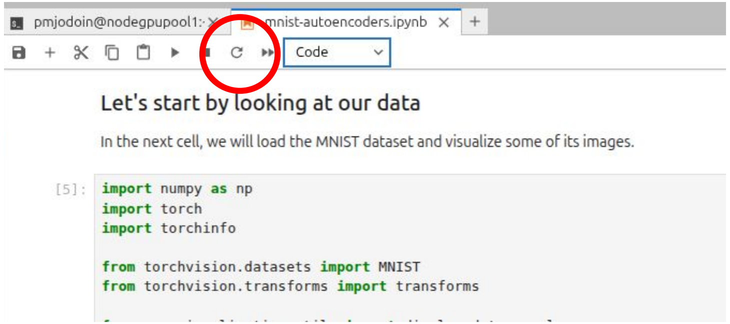
Click on **`mnist-autoencoders.ipynb`**. This will open the notebook.



If you see an error like this one:



Restart the kernel by clicking on the circular arrow and go back to the first cell

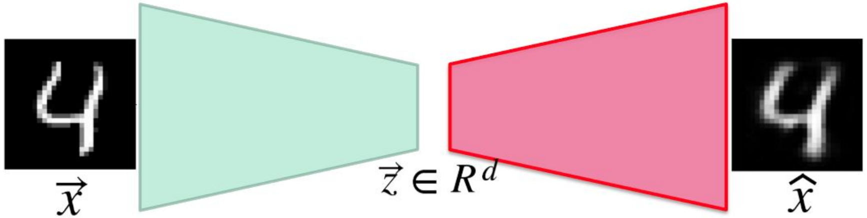


In this notebook you will learn…

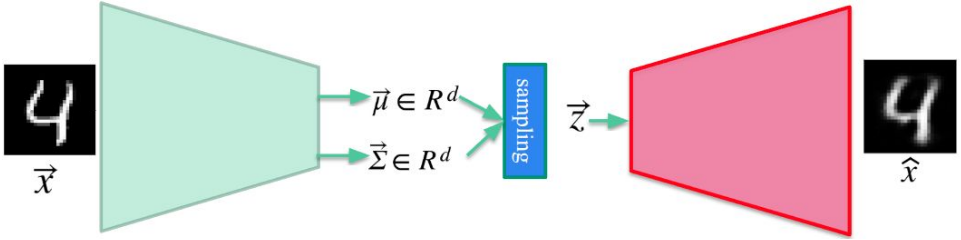
- What an Autoencoder is
- What a Variational Autoencoder is
- What a latent space is
- How you can use an Autoencoder to generate new MNIST data

In this notebook you will learn

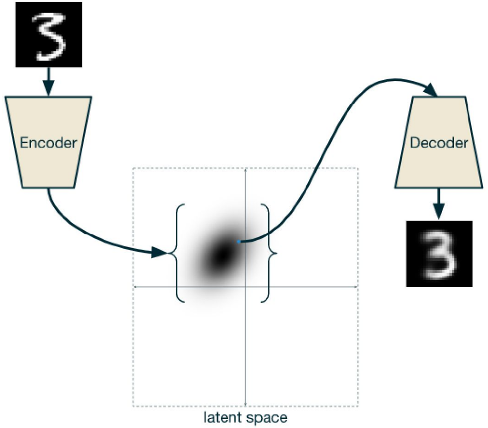
Autoencoder



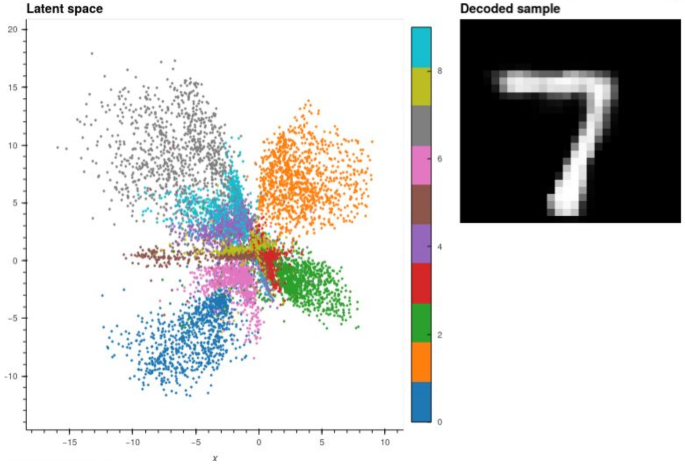
Variational Autoencoder



Latent space



Generative model

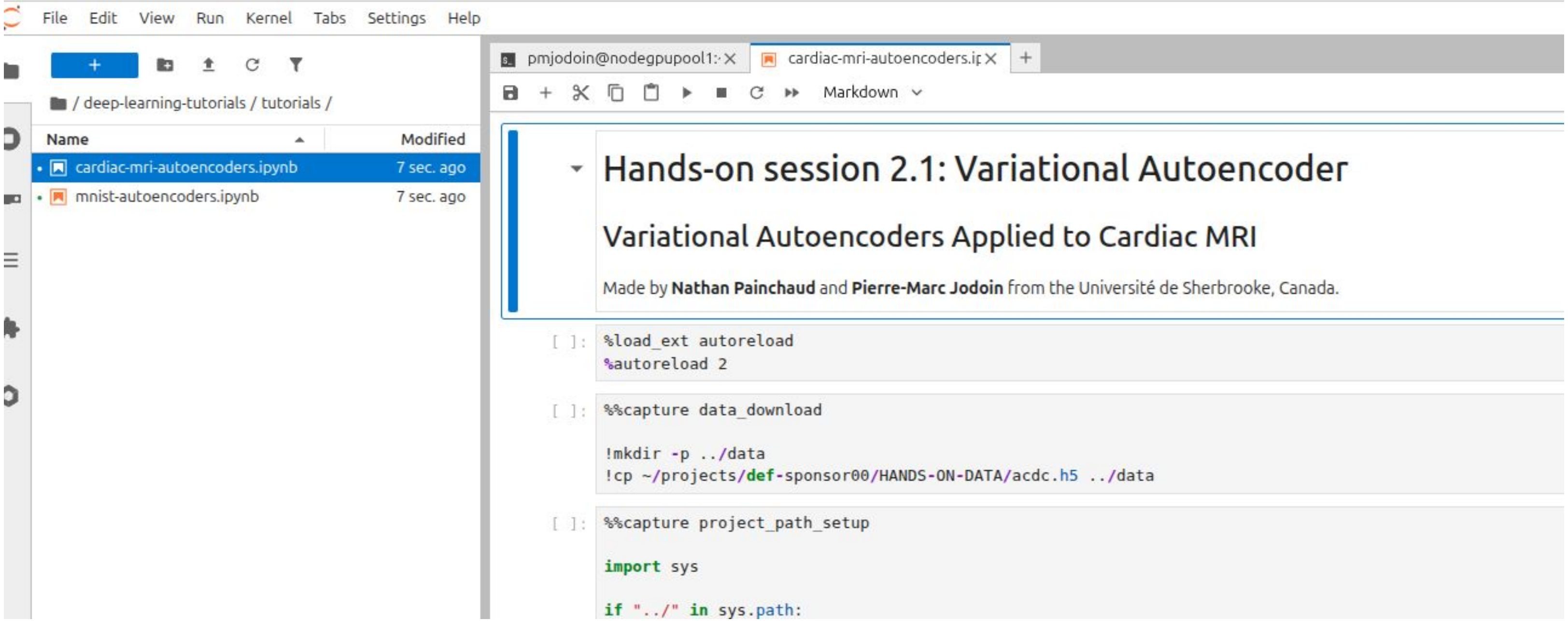


https://ijdykeman.github.io/ml/2016/12/21/cvae.html

The second notebook

Using the menu on the left, navigate to **deep-learning-tutorials/tutorials**.

Click on **`cardiac-mri-autoencoders.ipynb`**. This will open the notebook.



On the second notebook

In this notebook, you will learn how to

- Apply an Autoencoder to medical imaging data
- Learn a cardiac shape latent space
- Generate synthetic cardiac shapes

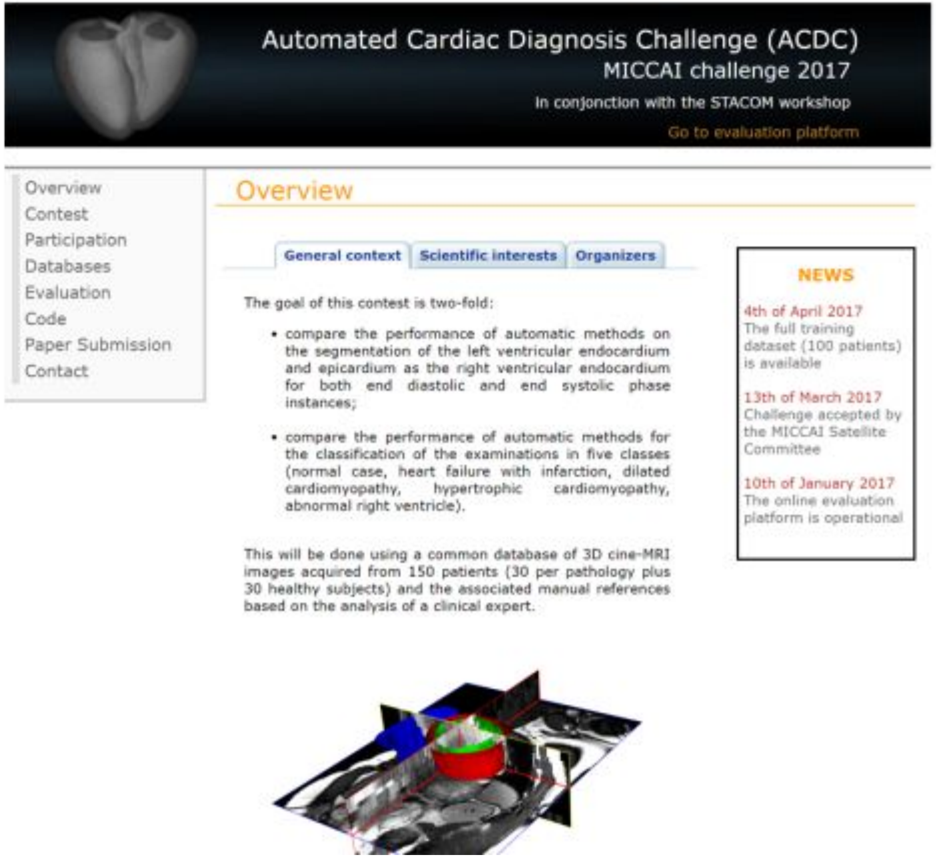
MICCAI 2017

Automated Cardiac Diagnosis Challenge

150 exams (all from different patients) divided into 5 groups:

- dilated cardiomyopathy (DCM),
- hypertrophic cardiomyopathy (HCM),
- myocardial infarction (MINF),
- abnormal right ventricle (RV)
- patients without cardiac disease (NOR).

website: ***creatis.insa-lyon.fr/Challenge/acdc***



Automated Cardiac Diagnosis Challenge (ACDC)
MICCAI challenge 2017
In conjonction with the STACOM workshop
Go to evaluation platform

- Overview
- Contest
- Participation
- Databases
- Evaluation
- Code
- Paper Submission
- Contact

Overview

General context | Scientific interests | Organizers

The goal of this contest is two-fold:

- compare the performance of automatic methods on the segmentation of the left ventricular endocardium and epicardium as the right ventricular endocardium for both end diastolic and end systolic phase instances;
- compare the performance of automatic methods for the classification of the examinations in five classes (normal case, heart failure with infarction, dilated cardiomyopathy, hypertrophic cardiomyopathy, abnormal right ventricle).

This will be done using a common database of 3D cine-MRI images acquired from 150 patients (30 per pathology plus 30 healthy subjects) and the associated manual references based on the analysis of a clinical expert.

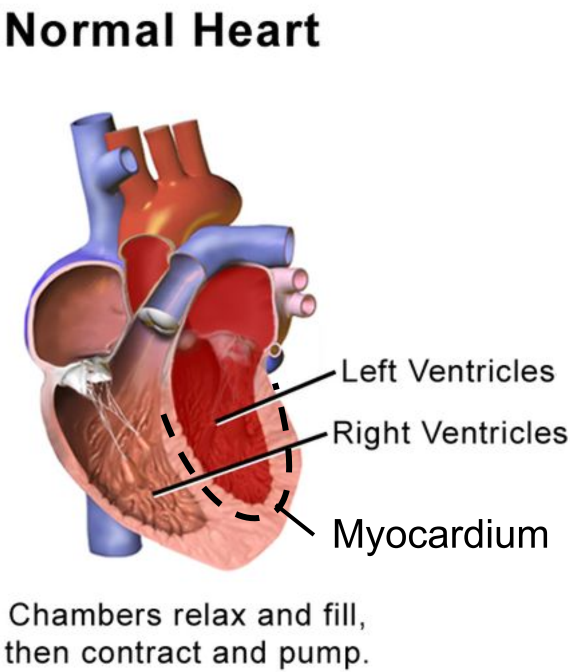
NEWS

4th of April 2017
The full training dataset (100 patients) is available

13th of March 2017
Challenge accepted by the MICCAI Satellite Committee

10th of January 2017
The online evaluation platform is operational

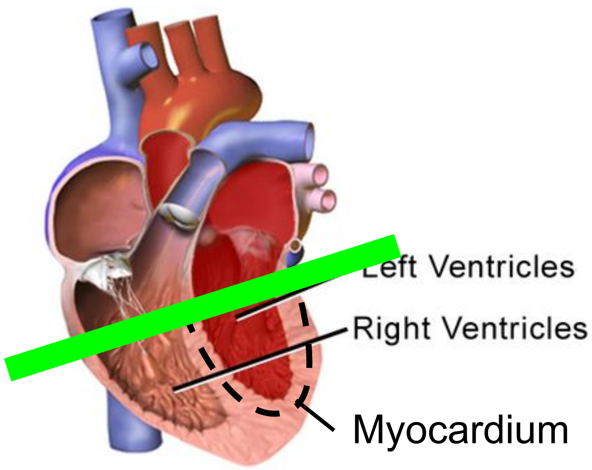
Cardiac anatomy



https://commons.wikimedia.org/

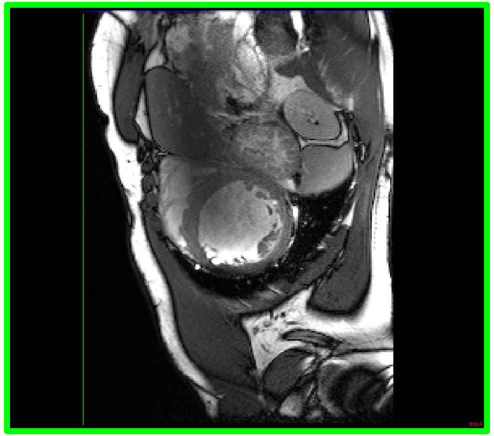
Cardiac anatomy

Normal Heart



Chambers relax and fill,
then contract and pump.

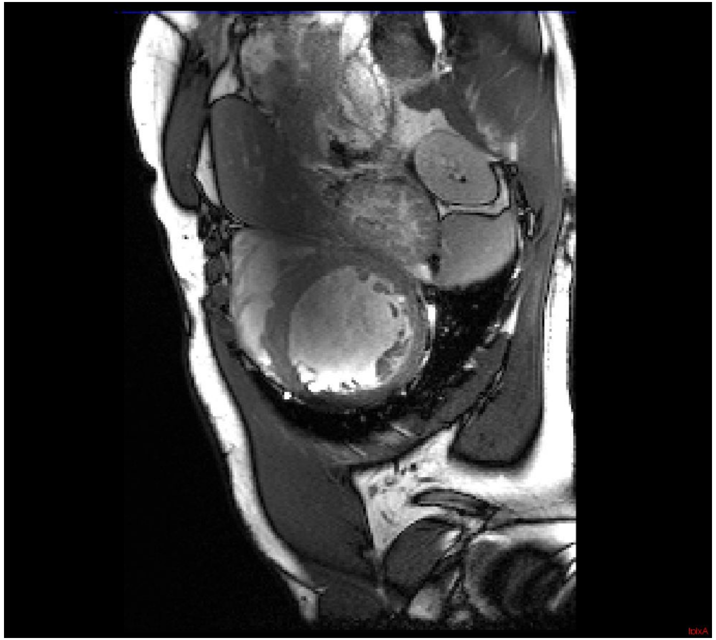
Cross-section : **short axis view**



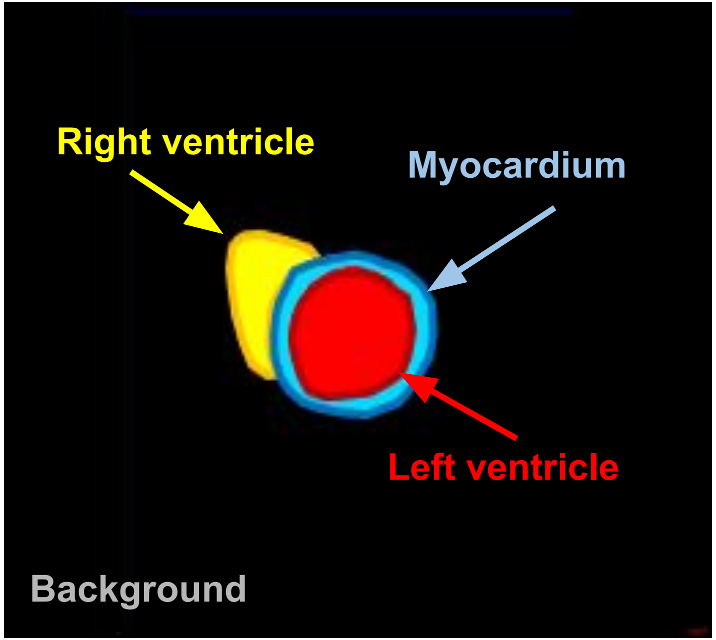
https://commons.wikimedia.org/

ACDC

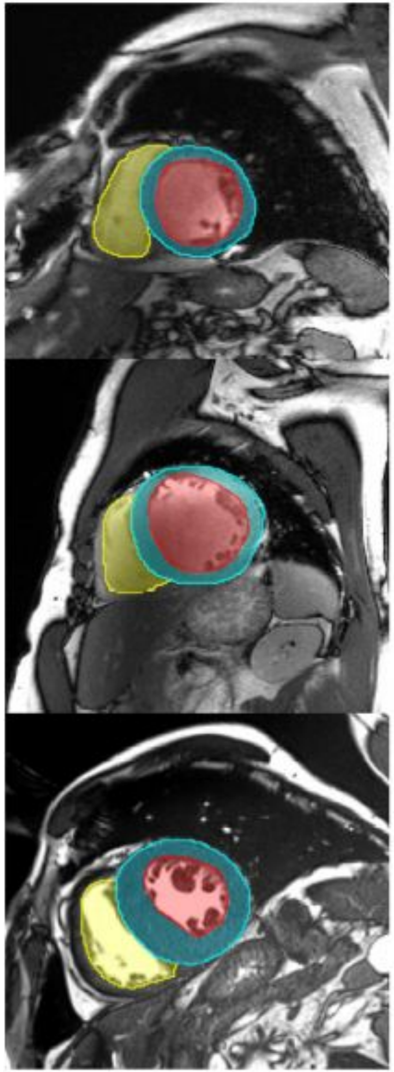
Short axis cardiac MRI



Short axis segmentation map



Other examples:



ACDC

Convolutional Variational autoencoder